7 CURRENT AND VOLTAGE IN CIRCUITS

We rely on electricity in many areas of our lives. This chapter looks at what electric current is. You will learn what happens to electric current in different circuits, and what effect it has.

Look around the room you are in. If you are at home, you will probably be able to see a television, a radio or a computer. If you are in a science laboratory, you may be able to see a projector, a power supply or lights in the ceiling. These and many other everyday objects need electric currents if they are to work. But what are electric currents? How are they produced and what do they do in a circuit?



▲ Figure 7.1 The gas in this tube glows when there is a current.

LEARNING OBJECTIVES

- Explain why a series or parallel circuit is more appropriate for particular applications, including domestic lighting
- Know that electric current in solid metallic conductors is a flow of negatively charged electrons
- Know and use the relationship between voltage, current and resistance:

  voltage = current × resistance

  $V = I \times R$
- Know that current is the rate of flow of charge
- Know and use the relationship between charge, current and time:

  charge = current × time

  $Q = I \times t$
- Know that lamps and LEDs can be used to indicate the presence of a current in a circuit
- Know that the voltage across two components connected in parallel is the same
- Know that voltage is the energy transferred per unit charge passed
- Know and use the relationship between energy transferred, charge and voltage:

  energy transferred = charge × voltage

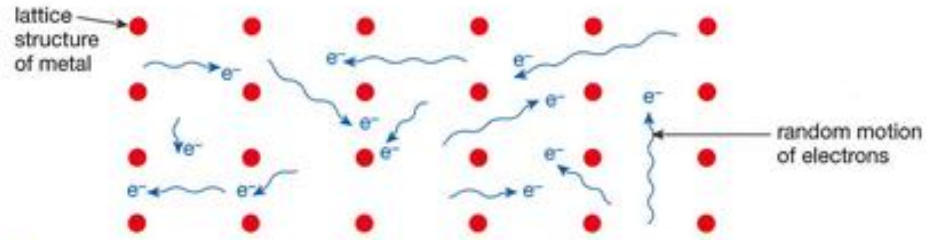
  $E = Q \times V$
- Know that the volt is a joule per coulomb
- Understand why current is conserved at a junction in a circuit

CONDUCTORS, INSULATORS AND ELECTRIC CURRENT

An electric current is a flow of charge. In metal wires the charges are carried by very small particles called electrons.

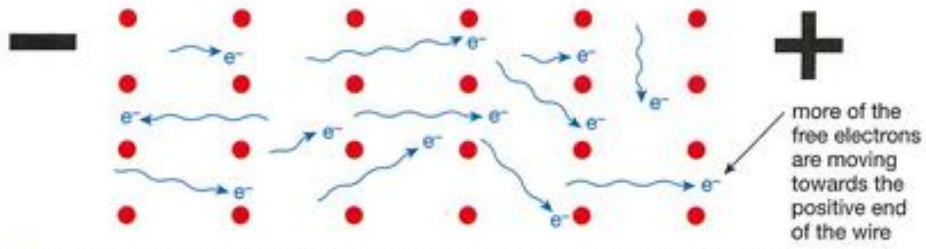
Electrons flow easily through all metals. We therefore describe metals as being good conductors of electricity. Electrons do not flow easily through plastics – they are poor conductors of electricity. A very poor conductor is known as an insulator and is often used in situations where we want to prevent the flow of charge – for example, in the casing of a plug.

In metals, some electrons are free to move between the **atoms**. Under normal circumstances this movement is random – that is, the number of electrons flowing in any one direction is roughly equal to the number flowing in the opposite direction. There is therefore no overall flow of charge.



▲ Figure 7.2a With no voltage there is an equal flow of electrons in all directions.

If, however, a cell or battery is connected across the conductor, more of the electrons now flow in the direction away from the negative terminal and towards the positive terminal than in the opposite direction. We say 'there is now a net flow of charge'. This flow of charge is what we call an electric current.



▲ Figure 7.2b When a voltage is applied more electrons will move towards the positive.

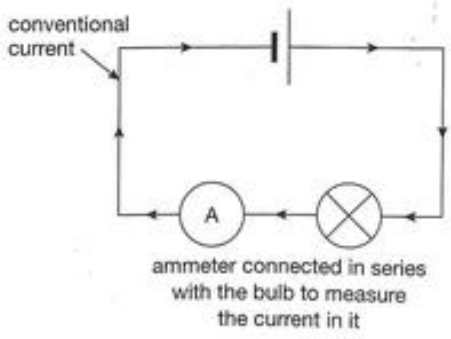
In insulators, all the electrons are held tightly in position and are unable to move from atom to atom. Charges are therefore unable to move through insulators.

EXTENSION WORK

When scientists first experimented with charges flowing through wires, they assumed that it was positive charges that were moving and that current travels from the positive to the negative. We now know that this is incorrect and that when an electric current passes through a wire it is the negative charges or electrons that move. Nevertheless when dealing with topics such as circuits and motors, it is still considered that electrons flow from positive to negative. This is conventional current.

MEASURING CURRENT





▲ Figure 7.3 An ammeter is used to measure current in a circuit. It has a very low resistance and so has almost no effect on the current.

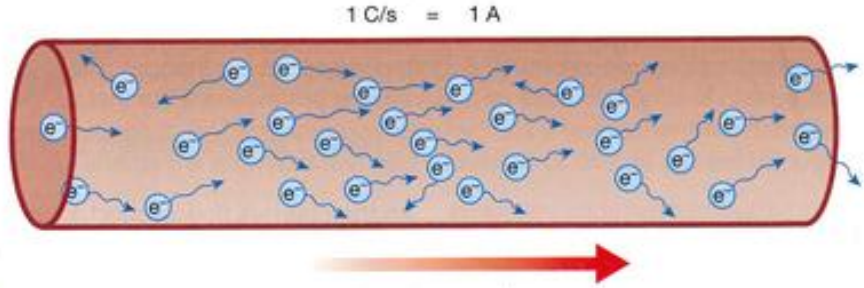
We measure the size of the current in a circuit using an ammeter. The ammeter is connected in series (see page 72) with the part of the circuit we are interested in.

The size of an electric current indicates the rate at which the charge flows.

The charge carried by one electron is very small and would not be a very useful measure of charge in everyday life. It would be a little like asking how far away the Moon is from the Earth ... and getting the answer in mm!

To avoid this problem we measure electric charge (Q) in much bigger units called coulombs (C). One coulomb of charge is equal to the charge carried by approximately six million, million, million (6×10^{18}) electrons.

We measure electric current (I) in amperes or amps (A). If there is a current of 1 A in a wire it means that 1 C of charge is passing along the wire each second.



▲ Figure 7.4 One coulomb of charge flowing each second is one amp.

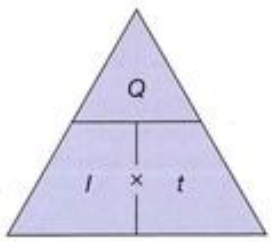
We can calculate the total charge that passes along a wire using the equation:

charge, Q (coulombs) = current, I (amps) × time, t (seconds)

$$Q = I \times t$$

HINT

If an examination question asks you to write out the equation for calculating charge, current or time, always give the actual equation such as $Q = I \times t$. You may not get the mark if you just draw the triangle.



▲ Figure 7.5 You can use the triangle method for rearranging equations like this.

EXAMPLE 3

a Calculate the charge flowing through a wire in 5 s if the current is 3 A.

$Q = I \times t$

$= 3\text{ A} \times 5\text{ s}$

$= 15\text{ C}$

b How many electrons flow through the wire in this time?

1 C of charge is carried by 6×10^{18} electrons

15 C of charge requires $15 \times 6 \times 10^{18}$ electrons = 90×10^{18} electrons

So 90 million, million, million electrons will flow along the wire in 5 s.

VOLTAGE

We often use cells or batteries to move charges around circuits. We can imagine them as being 'electron pumps'. They transfer energy to the charges. The amount of energy given to the charges by a cell or battery is measured in volts (V) and is usually indicated on the side of the battery or cell.

If we connect a 1.5 V cell into a circuit (Figure 7.6) and current flows, 1.5 J of energy is given to each coulomb of charge that passes through the cell.

If two 1.5 V cells are connected in series (Figure 7.7) so that they are pumping (pushing) in the same direction, each coulomb of charge will receive 3 J of energy.

The volt is a joule per coulomb.



▲ Figure 7.6 When one coulomb of charge passes through this cell it gains 1.5 J of energy.



▲ Figure 7.7 When one coulomb of charge passes through both these cells in turn it gains 3 J of energy.

KEY POINT

When several cells are connected together it is called a battery.

KEY POINT

Cells and batteries provide current which moves in one direction. This is known as direct current (d.c.).

As the charges flow around a circuit the energy they carry is transferred by the components they pass through. For example, when current passes through a bulb, energy is transferred to the surroundings as heat and light. When a current passes through the speaker of a radio, most of the energy is transferred as sound.

In the external part of a circuit (outside the cell or battery) the voltage across each component tells us how much energy it is transferring. If the voltage across a component is 1 V this means that the component is transferring 1 J of energy each time 1 C of charge passes through it.

We can describe the relationship between the energy transferred, charge and voltage using the equation:

energy transferred, E (joules) = charge, Q (coulombs) × voltage, V (volts)

$$E = Q \times V$$

EXAMPLE 2

The voltage across a light bulb is 12 V. Calculate the electrical energy transferred when 50 C of charge passes through it.

$E = Q \times V$

$= 50\ \text{C} \times 12\ \text{V}$

$= 600\ \text{J}$

Because $Q = I \times t$, we can also write this equation as $E = I \times t \times V$.

EXAMPLE 3

The voltage across a heater is 230 V. There is a current of 5 A through the heater for 5 mins. Calculate the total amount of energy transferred during this time.

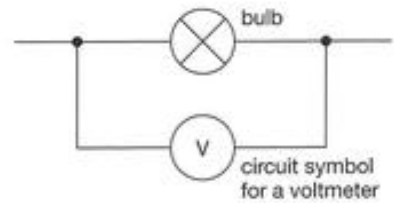
$E = I \times t \times V$

$= 5\ \text{A} \times (5 \times 60)\ \text{s} \times 230\ \text{V}$

$= 345\,000\ \text{J}$ or 345 kJ

MEASURING VOLTAGES





▲ Figure 7.8 A voltmeter measures voltages across a component.

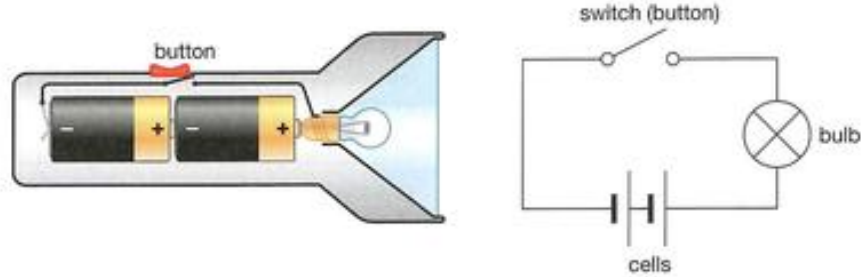
KEY POINT

A voltmeter has a very high resistance, so very little current flows through it.

We measure voltages using a voltmeter. This is connected across (in parallel with) the component we are investigating. A voltmeter connected across a cell or battery will measure the energy given to each coulomb of charge that passes through it. A voltmeter connected across a component will measure the electrical energy transferred when each coulomb of charge passes through it.

ELECTRICAL CIRCUITS

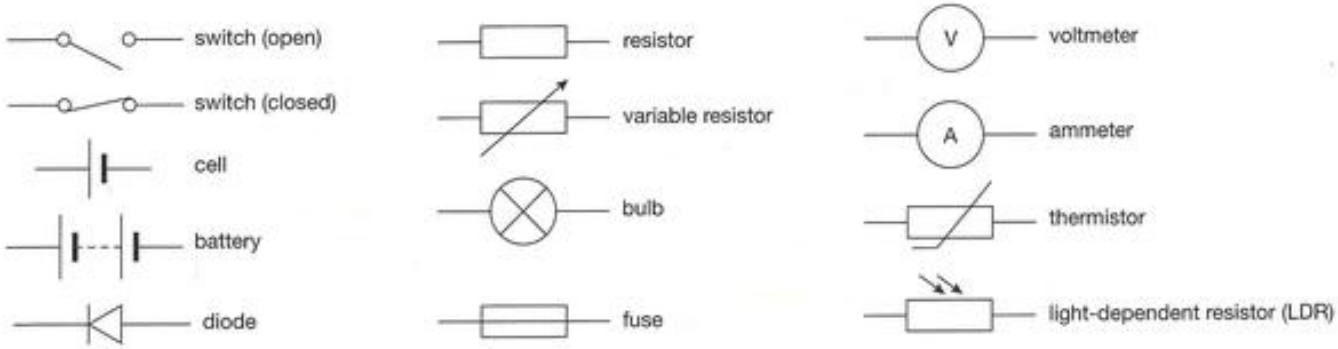
When the button on the torch shown in Figure 7.9 is pressed, the circuit is complete – that is, there are no gaps. Charges are able to flow around the circuit and the torch bulb glows. When the button is released the circuit becomes incomplete. Charges cease to flow and the bulb goes out.



▲ Figure 7.9 A torch contains a simple electrical circuit – a series circuit.

KEY POINT

Drawing diagrams of the actual components in a circuit is a very time-consuming and skilful task. It is much easier to use symbols for each of the components. Diagrams drawn in this way are called circuit diagrams. Figure 7.10 shows common circuit components and their symbols. You should know the common symbols but the less common ones will be given to you in the exam if you need them. Do not waste time memorising the less common ones.



▲ Figure 7.10 Circuit symbols



▲ Figure 7.11 Glowing LEDs indicate which circuits are working

IS IT ON?

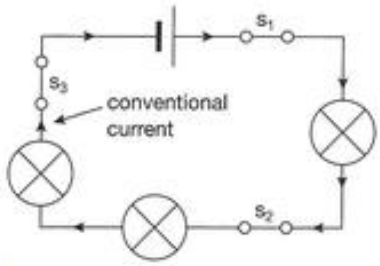
We sometimes put a small bulb or lamp in a circuit to show us if a circuit is 'turned on'. When there is a current in the circuit the bulb glows or shines. Light emitting diodes (LEDs) also glow when there is a current in a circuit but they require far less energy than bulbs. This is why many appliances such as TVs, DVD players and routers use small LEDs to show when the appliance is working or on standby.

SERIES AND PARALLEL CIRCUITS

There are two main types of electrical circuit. There are those circuits where there are no branches or junctions and there is only one path the current can follow. These simple 'single loop' circuits are called series circuits.

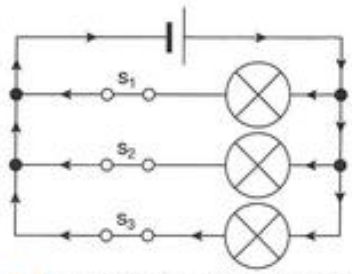
Circuits that have branches or junctions and more than one path that the current can follow are called parallel circuits.

a



▲ Figure 7.12a A typical series circuit. Opening any one of these switches will turn all three bulbs off.

b



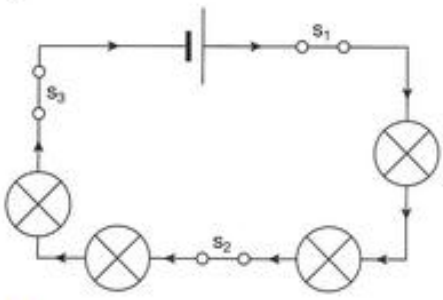
▲ Figure 7.12b A typical parallel circuit. Opening any one of these switches will turn off just the bulb in that part of the circuit.

Series circuits and parallel circuits behave differently. This makes them useful in different situations.

In a series circuit containing bulbs:

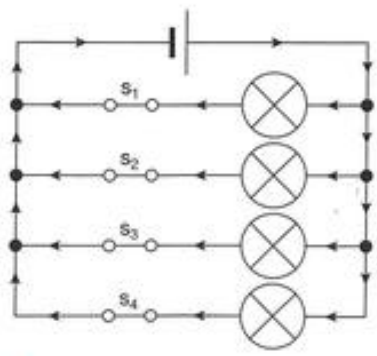
- One switch placed anywhere in the circuit can turn all the bulbs on and off.
- If any one of the bulbs breaks, it causes a gap in the circuit and all of the other bulbs will 'stop working'.
- The energy supplied by the cell is 'shared' between all the bulbs, so the more bulbs you add to a series circuit the less bright they all become.

a



▲ Figure 7.13a Adding an extra bulb in series will result in the bulbs shining less brightly.

b



▲ Figure 7.13b Adding an extra bulb in parallel does not affect the brightness of the other bulbs.

In a parallel circuit containing bulbs:

- Switches can be placed in different parts of the circuit to switch each bulb on and off individually, or all together.
- If one bulb breaks, only the bulbs on the same branch of the circuit will be affected.
- Each branch of the circuit receives the same voltage, so if more bulbs are added to a circuit in parallel they all keep the same brightness.



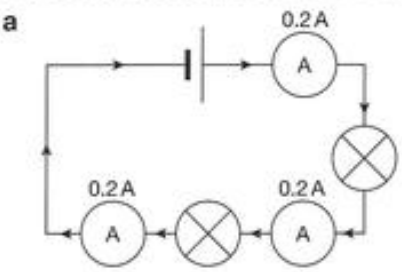
▲ Figure 7.14 Decorative lights are usually wired in series.

Decorative lights are often wired in series. Each bulb only needs a low voltage, so even when the voltage from the mains supply is 'shared' out between them each bulb still gets enough energy to produce light. Unfortunately, if the filament in one of the bulbs breaks then all the other bulbs will go out.

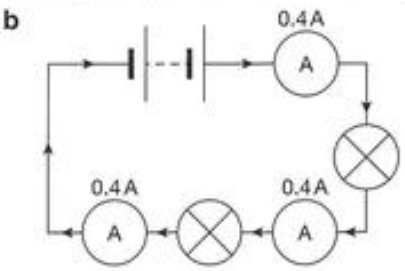
The lights in your home are wired in parallel. We know this because lights can be switched on and off separately, and the brightness of each light does not change when other lights are on or off. Also, if a bulb breaks or is removed, you can still use the other lights.

CURRENT IN A SERIES CIRCUIT

In a series circuit the current is the same in all parts; current is not used up.

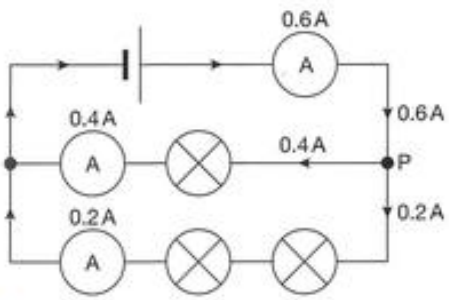


▲ Figure 7.15a In a series circuit the current does not vary.



▲ Figure 7.15b The addition of a second cell doubles the voltage applied to the circuit so the current will also double.

The size of the current in a series circuit depends on the voltage supplied to it, and the number and type of the other components in the circuit. If a second identical cell is added in series the voltage will double and so the current will also double.



▲ Figure 7.16 There are different currents in the different parts of a parallel circuit.

CURRENT IN A PARALLEL CIRCUIT

In a parallel circuit the currents will not be the same in different parts of the circuit. The types of components in each of the different parts will affect the currents.

In a parallel circuit the number of electrons that flow into a junction each second must be equal to the number that leave each second. This means that the currents entering a junction must always be equal to those that leave. For example, in Figure 7.16 the current that enters junction P is 0.6 A. The current that leaves is 0.4 A + 0.2 A = 0.6 A.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING

More questions on electrical circuits can be found at the end of Unit 2 on page 93.

1 Current is a flow of charge.

a What are the charge carriers in metals?

b Explain why charges are able to flow through metals but not through a plastic.

SKILLS PROBLEM SOLVING

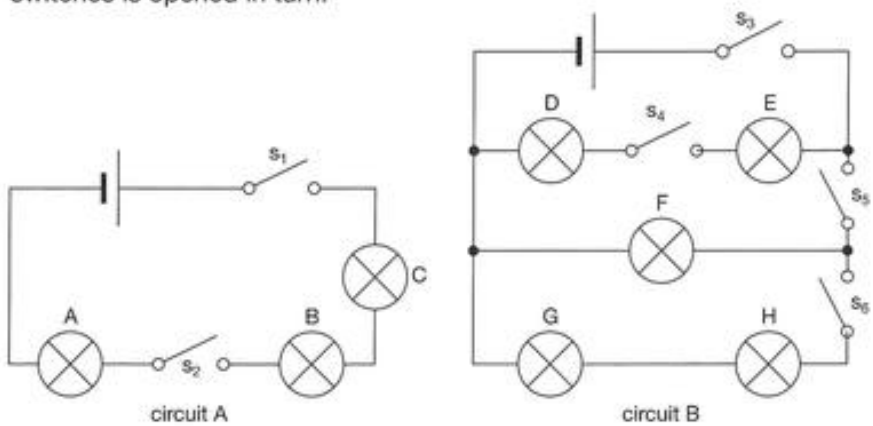
c If the current in a heater is 3 A, calculate the charge that flows through it in:
i 1 s
ii 10 min
iii 1 hour.

SKILLS CRITICAL THINKING

2 a Explain the differences between:
i a complete circuit and an incomplete circuit
ii a series circuit and a parallel circuit.

SKILLS ANALYSIS

b Look carefully at the circuits shown below. Assuming that all switches are initially closed, decide which of the bulbs go out when each of the switches is opened in turn.



c In circuit A, which bulb(s) glow the brightest when all the switches in the circuit are closed?

SKILLS CRITICAL THINKING

d Explain your answer to part c.

SKILLS INTERPRETATION

3 The voltage between two points in a circuit is measured using a voltmeter.

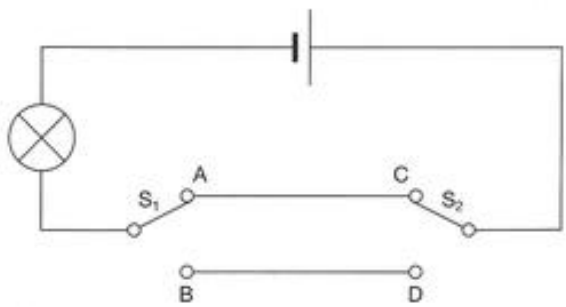
a Draw a circuit diagram to show how a voltmeter should be connected to measure:
i the voltage across a bulb ii the voltage of a cell.

SKILLS CRITICAL THINKING

b Explain in your own words the phrase 'a cell has a voltage of 1.5 V'.

SKILLS REASONING

4 The diagram below shows a circuit containing two 2-way switches.



a Explain in your own words what happens when each of the switches is moved to a new position.

b Suggest one important application of this circuit in the home.

5 Should the lights for the main street in a town be wired in series or in parallel? Explain your answer.

6 Why would it not be a good idea to connect all the different parts of an electric cooker (oven, grill, heating plates) in series?